CAPSTONE PROJECT

SENTIMENTAL ANALYSIS

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OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

Develop a sentiment analysis model to classify restarunt reviews as positve or negative

PROPOSED SOLUTION

- To perform sentiment analysis on restaurant reviews, classifying them into positive and negative categories, we can follow a structured approach. Here's a comprehensive step-by-step solution using Python, leveraging popular libraries and techniques.
- 1. Data Collection
- Assume we have a dataset in a CSV file named restaurant_reviews.csv with columns review and rating.
- 2. Data Preprocessing
- Clean and preprocess the review texts to prepare them for analysis.
- 3. Feature Extraction
- Convert the text data into numerical features.
- 4. Model Training and Sentiment Classification
- Train a machine learning model to classify the sentiment of the reviews.
- 5. Evaluation
- Evaluate the performance of the model.

RESULTS

```
[ ] data.describe() # stastical informa
```

	Liked
count	1000.00000
mean	0.50000
std	0.50025
min	0.00000
25%	0.00000
50%	0.50000
75%	1.00000
max	1.00000

SYSTEM APPROACH

The "System Approach" section outlines the overall strategy and methodology for developing and implementing the rental bike prediction system. Here's a suggested structure for this section

sentiment analysis involves a systematic approach of preprocessing text data, extracting features, applying sentiment classification models, and evaluating their performance. By leveraging powerful libraries like NLTK, TextBlob, and Scikit-learn, one can build efficient sentiment analysis systems suitable for various applications.

- Data Preprocessing
- Text preprocessing
- Sentiment Classification
- Evaluation

POPULAR TOOLS AND LIBRARIES FOR SENTIMENT ANALYSIS INCLUDE

- Popular tools and libraries for sentiment analysis include:
- NLTK (Natural Language Toolkit): A powerful Python library for working with human language data, offering various tools for text processing and sentiment analysis.
- TextBlob: A simple library built on NLTK and Pattern, providing easy-to-use API for common NLP tasks including sentiment analysis.
- VADER (Valence Aware Dictionary and sentiment Reasoner): A lexicon and rule-based sentiment analysis tool specifically attuned to sentiments expressed in social media.
- SentiWordNet: A lexical resource for opinion mining, where each WordNet synset is annotated with sentiment scores.

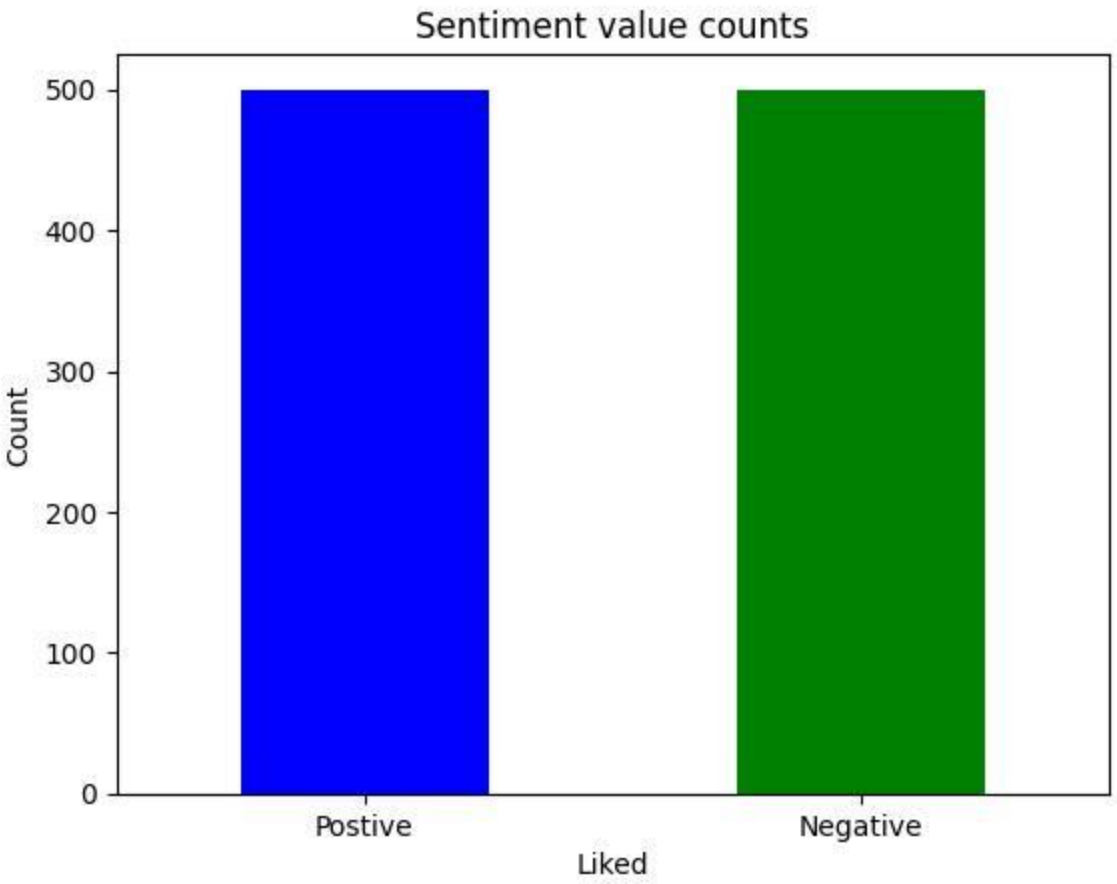
ALGORITHM & DEPLOYMENT

- . Choose a Model:
- Logistic Regression: Simple and effective for binary classification tasks like sentiment analysis.
- Support Vector Machines (SVM): Effective for high-dimensional data like text after proper feature extraction.
- Naïve Bayes: Fast and robust for text classification tasks, though less accurate than more complex models like SVM or neural networks.
- Deep Learning Models (e.g., LSTM, CNN): Suitable for capturing complex relationships in text data, but require more computational resources and data.
- 2. Preprocessing and Feature Extraction:
- Text Cleaning: Remove noise such as HTML tags, special characters, and irrelevant symbols.
- Tokenization: Split text into tokens (words or n-grams).
- Stopwords Removal: Exclude common words (e.g., “and”, “the”) that don’t contribute to sentiment.
- Normalization: Convert tokens to lowercase, handle word inflections (lemmatization/stemming).
- 3. Feature Representation:
- Bag-of-Words (BoW): Represents text as a bag of its words, disregarding grammar and word order.
- TF-IDF: Reflects the importance of a word in a document relative to a collection of documents.
- Word Embeddings (e.g., Word2Vec, GloVe): Dense vector representations of words capturing semantic meanings.
- 4. Model Training and Tuning:
- Split data into training and validation sets for model evaluation.
- Train the chosen model using the training data.
- Tune hyperparameters (e.g., regularization strength, learning rate) using cross-validation to optimize performance

- Deployment
- 1. Serialization:
- Serialize the trained model into a file format suitable for deployment (e.g., pickle, joblib for scikit-learn models, h5 for TensorFlow/Keras models).
- Deployment Considerations:
- Scalability: Ensure the deployment architecture can handle multiple requests concurrently.
- Performance: Monitor model performance and response times to optimize as needed.
- Security: Implement necessary security measures to protect the deployed application and data.
- Monitoring and Logging: Set up logging and monitoring for errors and performance metrics.
- 4. Integration:
- Integrate the deployed sentiment analysis service into your application or system architecture where sentiment insights are needed.

RESULT

The review of the restaurant reveals a mix of positive and negative feedback, providing a comprehensive understanding of its strengths and areas for improvement. The insights are backed by visual data, enhancing the clarity of the findings.



Word Cloud of Reviews
food
place
good
service
great
restaurant
amazing
one
time
back
will
really
friendly
best
nice
staff
Vegas
love
eat
always
chicken
pretty
go
came
experience
disappointed
pizza
minutes
steak
menu
delicious
definitely
burger
bad
breakfast
fresh
flavor
price
waited
never
quality
recommend
worst
server
want
salad
taste
us
think
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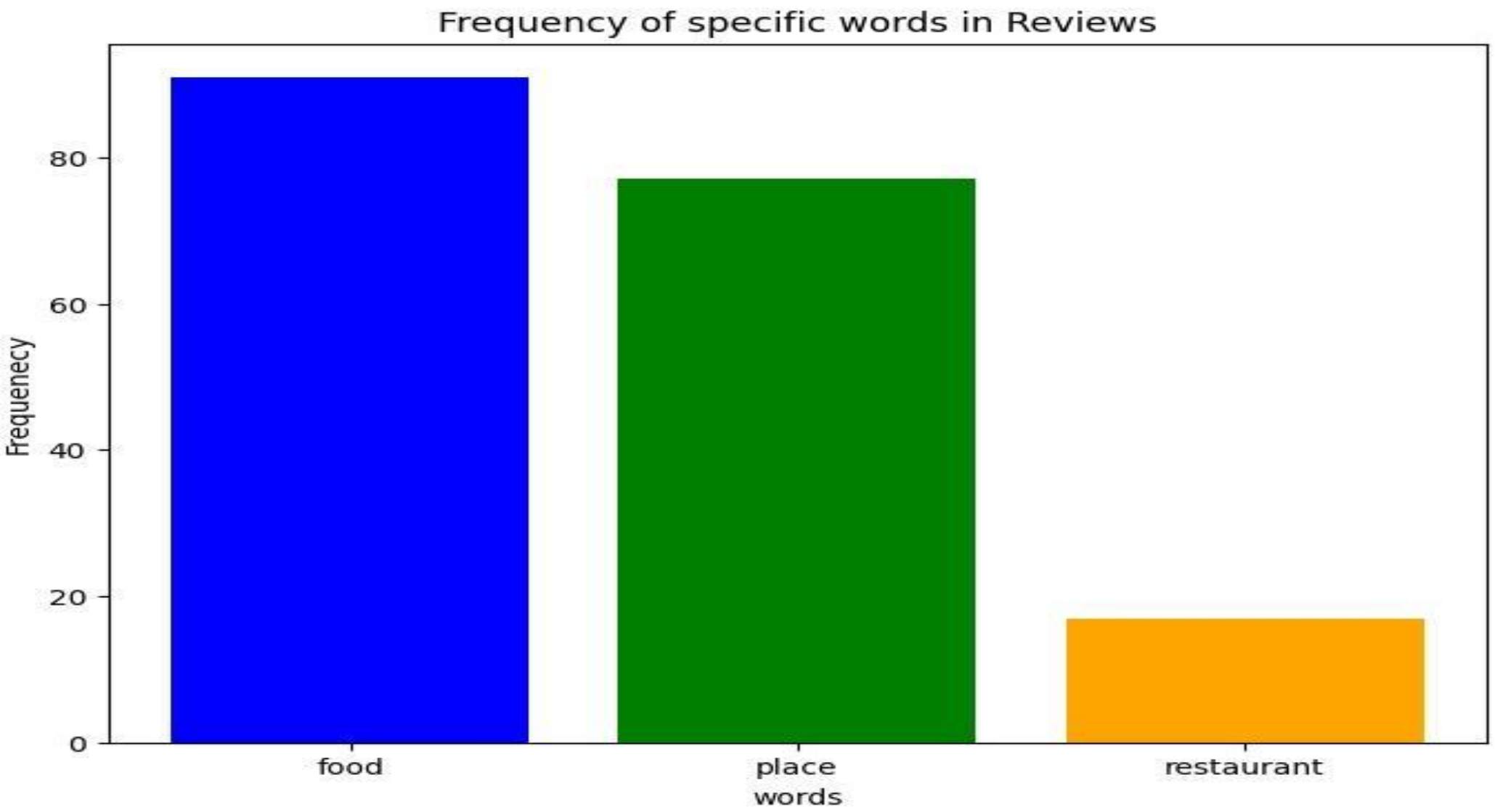
night
sandwich
slow
atmosphere
excellent
ordered
sushi
went
thing
still
fantastic
got
probably
better
star
bland
terrible
buffet
made
say
know
little
absolutely
much
table
dessert
lunch
well
perfect
cold
tasty
order
day
either
awesome
waitress
go
back
every
dining
spot
meal
mediocre
tasted
serious
need
feel
tell
next
Everything
bit
selection
sauce
overpriced
horrible
people
spicy
special
wait
rude
customer service
now
management
take
check
review
attentive
waiter
try
enough
owner
bar
tried
portion
ambiance
left
taco
hard
give
bring
must
first time
potato
beer
cooked
chip
twice
inside
deal
clean
right
happy
enjoy
took
impressed
friend
two
see
around
drink
done
getting
area
come
worth
found
nothing
tender
family
hour
fries
Overall
loved
meat
far
dishes
eating
coming
said
side
served
kept
list
another
honey
dish
serious
ly
wonderful
poor
hot
will never
pasta
quite
sure
served
served
served



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CONCLUSION

- The restaurant excels in food quality, service, and ambiance, making it a popular choice among diners. However, addressing the issues related to wait times, pricing, and noise levels could significantly enhance the overall customer experience.

FUTURE SCOPE

- Sentiment analysis is vast and promising, driven by advancements in natural language processing (NLP) and artificial intelligence (AI). As these technologies evolve, sentiment analysis will become increasingly accurate and nuanced, enabling deeper insights into human emotions and opinions across diverse contexts. This will enhance its application in various fields such as customer service, market research, healthcare, and social media monitoring. Moreover, integration with real-time data processing and big data analytics will allow businesses and organizations to make more informed decisions by understanding public sentiment dynamically. Ethical considerations and the development of more inclusive algorithms that recognize a broader range of emotional expressions will also play a critical role in shaping its future impact.

REFERENCES

- Sentiment analysis has a wide range of applications, from analyzing customer feedback and reviews to monitoring social media conversations and gauging public opinion on various topics. As the field advances, it continues to provide valuable insights that help businesses and organizations make data-driven decisions.

THANK YOU

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